

Question number	Answer	Mark
1(a)	C thylakoid membranes stroma The only correct answer is C. <i>A is incorrect because the light-dependent reactions take place in the thylakoid membranes</i> <i>B is incorrect because the light-dependent reactions take place in the thylakoid membranes and the light-independent reactions take place in the stroma</i> <i>D is incorrect because the light-independent reactions take place in the stroma</i>	(1)

Question number	Answer	Additional guidance	Mark
1(b)	An answer that includes three of the following points: - DNA (loop) drawn and labelled (1) - starch grain drawn and labelled (1) {envelope / inner membrane / outer membrane} drawn and labelled (1) - grana / grana stack / granum / (inter granal) lamellae (1) - ribosomes drawn and labelled (1)	IGNORE lipid droplets, stroma, thylakoid membranes ACCEPT plasmid / plasmid-like DNA ACCEPT starch granules ACCEPT / double membrane IGNORE size references	(3)

Question number	Answer	Mark
1(c)	The only correct answer is B. <i>A is incorrect because green wavelengths are reflected</i> <i>C is incorrect because green wavelengths are reflected</i> <i>D is incorrect because green wavelengths are reflected</i>	(1)

Question number	Answer	Additional guidance	Mark
1(d)	rate of photosynthesis at different wavelengths of light		(1)

Question number	Answer	Mark
1(e)(i)	The only correct answer is A <i>B is incorrect because dendrochronology is the study of tree growth rings</i> <i>C is incorrect because osmosis is the movement of free water molecules from a high solute potential to a lower solute potential</i> <i>D is incorrect because PCR amplifies the number of DNA molecules</i>	(1)

Question number	Answer	Mark
1(e)(ii)	The only correct answer is B. <i>A is incorrect because the R_f value of J is distance moved by J divided by distance moved by solvent front = $6 \div 7.5 = 0.800$</i> <i>C is incorrect because the R_f value of J is distance moved by J divided by distance moved by solvent front = $6 \div 7.5 = 0.800$</i> <i>D is incorrect because the R_f value of J is distance moved by J divided by distance moved by solvent front = $6 \div 7.5 = 0.800$</i>	(1)